

Closed-form metal line-cooling formulas in EXHALE

1. Overview and conventions

All metal line-cooling coefficients in `Cool_coeff.f90` are evaluated from closed-form analytic expressions fitted to the CHIANTI v11.0.2 database (Burgess–Tully descaled `.scups` effective collision strengths); only Fe I (1-D) and the density-dependent Fe II statistical-equilibrium coefficient (2-D) remain tabulated. Throughout, $\Lambda(T)$ denotes the effective cooling coefficient *per* ($n_e n_{\text{ion}}$) in $\text{erg cm}^3 \text{s}^{-1}$, in the optically thin limit (every collisional excitation is followed by a radiative decay); the cooling assembly multiplies by $\beta_{\text{esc}} n_e n_{\text{ion}}$. Lower levels are populated in Boltzmann equilibrium over the ground-term fine structure where one exists. Two practical conventions matter when reproducing the numbers: (i) the code uses $k_B = 1.38 \times 10^{-16} \text{ erg K}^{-1}$ (rounded, code-wide); (ii) the transition energies recovered by the fits are the CHIANTI `.scups` *theoretical* energies, which can differ from the observed-wavelength energies by a few percent (e.g. Mg II 4.277 vs. 4.43 eV) — using the observed value inflates the fit error from $\sim 1\%$ to $\sim 5\%$.

Fit scripts (in `cooling_data/`): `fit_cooling_formulas.py` (resonance lines, Fe II), `fit_cno_formulas.py` (C/N/O), `fit_fs_saturation.py` (fine-structure saturation). The companion notebooks `cooling_formula_fits.ipynb` and `cno_cooling_comparison.ipynb` reproduce every figure here.

2. Resonance lines: exact two-level skeleton

A single-channel resonance line obeys exactly

$$\Lambda(T) = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \Upsilon(T) \Delta E e^{-\Delta E/k_B T}, \quad (1)$$

so the only quantity to fit is the slowly varying effective collision strength, for which the Burgess–Tully type-1 asymptotic form

$$\Upsilon(T) = a + b \ln(1 + T/T_0) \quad (2)$$

suffices. Table 1 lists the fitted constants; Fig. 1 compares the formulas with the source tables. Na I D is reproduced essentially exactly because the underlying rate is itself a Van Regemorter expression with constant Υ (calibrated to CHIANTI Mg II/Ca II with $\bar{g} \simeq 0.2$; Na I has no CHIANTI electron-collision model).

Table 1: Resonance-line cooling: Eqs. (1–2). ΔE is the fitted (`.scups` theoretical) transition energy. “Max err” is the maximum deviation from the 201-point CHIANTI source table over 10^3 – 10^5 K.

channel	g_l	ΔE [eV]	a	b	T_0 [K]	max err
Mg I $\lambda 2853$	1	4.3381	0.355792	15.1223	94126.5	2.8%
Mg II h&k	2	4.2766	16.2305	20.3448	121896	1.1%
Ca II H&K	2	3.1438	14.7415	19.4920	36218.3	1.2%
Na I D	2	2.1037	36.0744	0 (const.)	—	0.01%

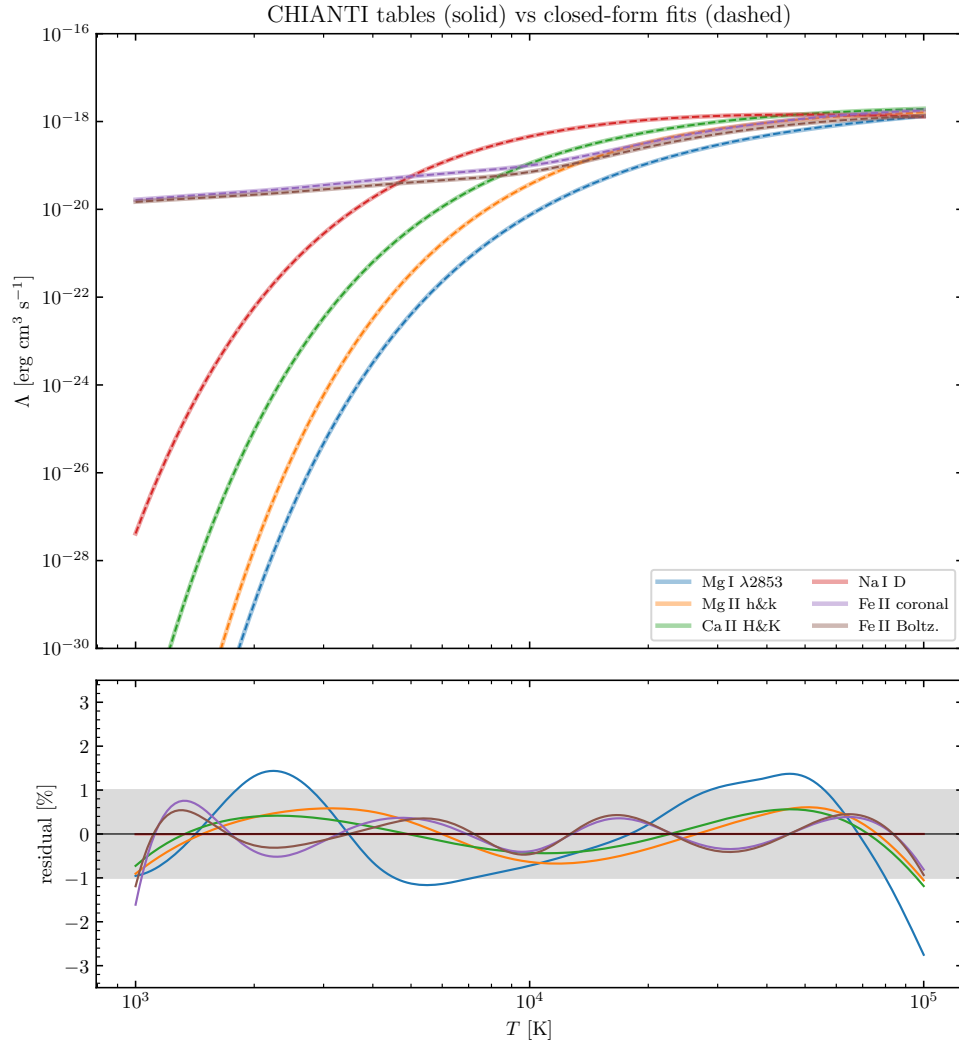


Figure 1: CHIANTI cooling tables (solid) vs. the closed-form fits (dashed) for the four resonance channels and the two Fe II population models (Sect. 3.). Bottom: signed residuals; the grey band is $\pm 1\%$.

3. Fe II: sum of effective two-level groups

Fe II cooling sums hundreds of transitions with ΔE from ~ 0.05 eV (a^6D fine structure) to several eV, so a single exponential cannot represent it. Four effective two-level groups do:

$$\Lambda(T) = T^{-1/2} \sum_{i=1}^4 A_i e^{-T_i/T}. \quad (3)$$

The fitted T_i (Table 2) map onto physically meaningful transition groups — infrared fine structure, optical metastable, UV resonance, and a high-energy tail (Fig. 2).

Table 2: Fe II coefficients for Eq. (3). “Coronal” populates the ground level only; “Boltzmann” distributes the lower levels over the metastable manifold ($E < 4.8$ eV, Huang 2023).

i	Fe II coronal (max err 1.6%)		Fe II Boltzmann (max err 1.2%)	
	A_i	T_i [K]	A_i	T_i [K]
1	1.9889×10^{-18}	1350.1	1.5019×10^{-18}	1175.7
2	1.6793×10^{-17}	9630.3	9.7419×10^{-18}	8678.5
3	6.5362×10^{-16}	58939.3	5.2485×10^{-16}	57594.7
4	7.4638×10^{-16}	139593.3	4.7387×10^{-16}	140620.2

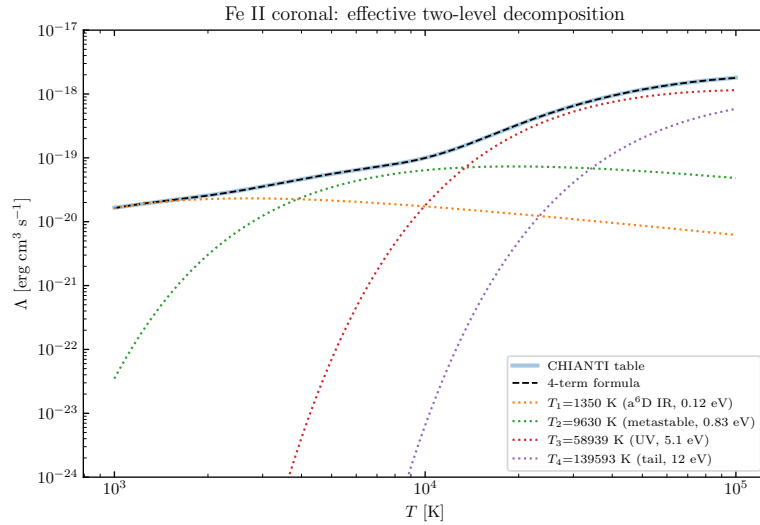


Figure 2: Fe II coronal cooling decomposed into the four effective two-level groups of Eq. (3).

In the cooling *assembly* the coronal formula is superseded by the 2-D statistical-equilibrium table $\Lambda_{\text{eff}}(T, n_e)$ (cool_logL_FeII_ne): the forbidden a^6D lines have $n_{\text{crit}} \sim 10^4$ – 10^7 cm $^{-3}$ and saturate at the dense base. The closed-form coronal expression is retained for the post-process scalar fallback and matches the low- n_e edge of the 2-D table to $\sim 1.6\%$.

4. C, N, O: CHIANTI fits vs. the legacy AIOLOS formulas

The C/N/O coolants inherited from AIOLOS (photochem.cpp; no in-code citation, likely Cloudy-calibrated) are

$$\begin{aligned}\Lambda_{\text{CI}}^{\text{AIO}} &= 10^{-24} + 3.1 \times 10^{-20} e^{-15162/T} [1 + (T/2 \times 10^4)^{1.5}], \\ \Lambda_{\text{CII}}^{\text{AIO}} &= 1.5 \times 10^{-23} + 3.1 \times 10^{-20} e^{-45162/T} [1 + (T/7500)^{1.5}], \\ \Lambda_{\text{OI}}^{\text{AIO}} &= 5.5 \times 10^{-24} + 1.1 \times 10^{-20} e^{-30162/T} [1 + (T/7500)^{0.5}], \\ \Lambda_{\text{OII}}^{\text{AIO}} &= 5.1 \times 10^{-20} e^{-35162/T} [1 + (T/7500)^{0.5}],\end{aligned}\tag{4}$$

with no nitrogen cooling at all. The replacement CHIANTI fits use the same multi-exponential form as Fe II, Eq. (3) with 4–6 terms (Table 3); lower levels are Boltzmann-populated over the ground-term fine structure (C I ³P, C II ²P, N II ³P, O I ³P; the ⁴S ground of N I and O II is a single level).

Figure 3 compares the three. The AIOLOS carbon fits track CHIANTI remarkably well near 10⁴ K (C I to ~1%), which suggests they were calibrated against data of similar provenance; the oxygen fits however deviate by 40–70% across the wind-launch region (1–3 × 10⁴ K), and nitrogen — a non-negligible coolant around 2 × 10⁴ K — was absent entirely. The runtime key `cno_cool` in `metals.inp` selects the source (1 = CHIANTI fits, the default; 0 = legacy AIOLOS).

Table 3: C/N/O coefficients for Eq. (3) (CHIANTI ground-term curves; per ($n_e n_{\text{ion}}$)). The notation $a(b)$ means $a \times 10^b$. Max err is vs. the recomputed CHIANTI curve over 10³–10⁵ K.

ion (max err)		$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$
C I (1.8%)	A_i	1.106(−20)	1.392(−18)	6.962(−18)	6.593(−17)	3.017(−16)	
	T_i [K]	2351	15172	25411	72609	177264	
C II (0.5%)	A_i	3.004(−20)	2.410(−20)	3.904(−17)	3.332(−16)	1.360(−15)	
	T_i [K]	61.4	7183	64104	124809	246136	
N I (0.6%)	A_i	1.514(−18)	4.877(−18)	1.160(−17)	1.575(−16)	3.175(−16)	
	T_i [K]	29686	34694	50741	139916	254192	
N II (0.1%)	A_i	2.406(−20)	1.244(−20)	8.040(−18)	1.545(−17)	2.884(−16)	6.944(−16)
	T_i [K]	49.9	6521	23771	64487	157103	295894
O I (2.1%)	A_i	3.236(−22)	6.572(−20)	1.803(−18)	6.762(−18)	3.372(−17)	
	T_i [K]	848	19059	31568	70528	178827	
O II (0.2%)	A_i	1.768(−17)	8.071(−18)	2.344(−16)	8.789(−16)		
	T_i [K]	44129	59732	179091	325669		

5. Density-dependent fine-structure saturation

The coronal C II and O I curves include the ground-term fine-structure floors ([C II] 158 μm , [O I] 63 μm) *unsaturated*, although their critical densities ($n_{\text{crit,e}} \simeq 20 \text{ cm}^{-3}$ for [C II]; $n_{\text{crit,H}} \simeq 4 \times 10^5 \text{ cm}^{-3}$ for [O I]) lie far below the atmosphere

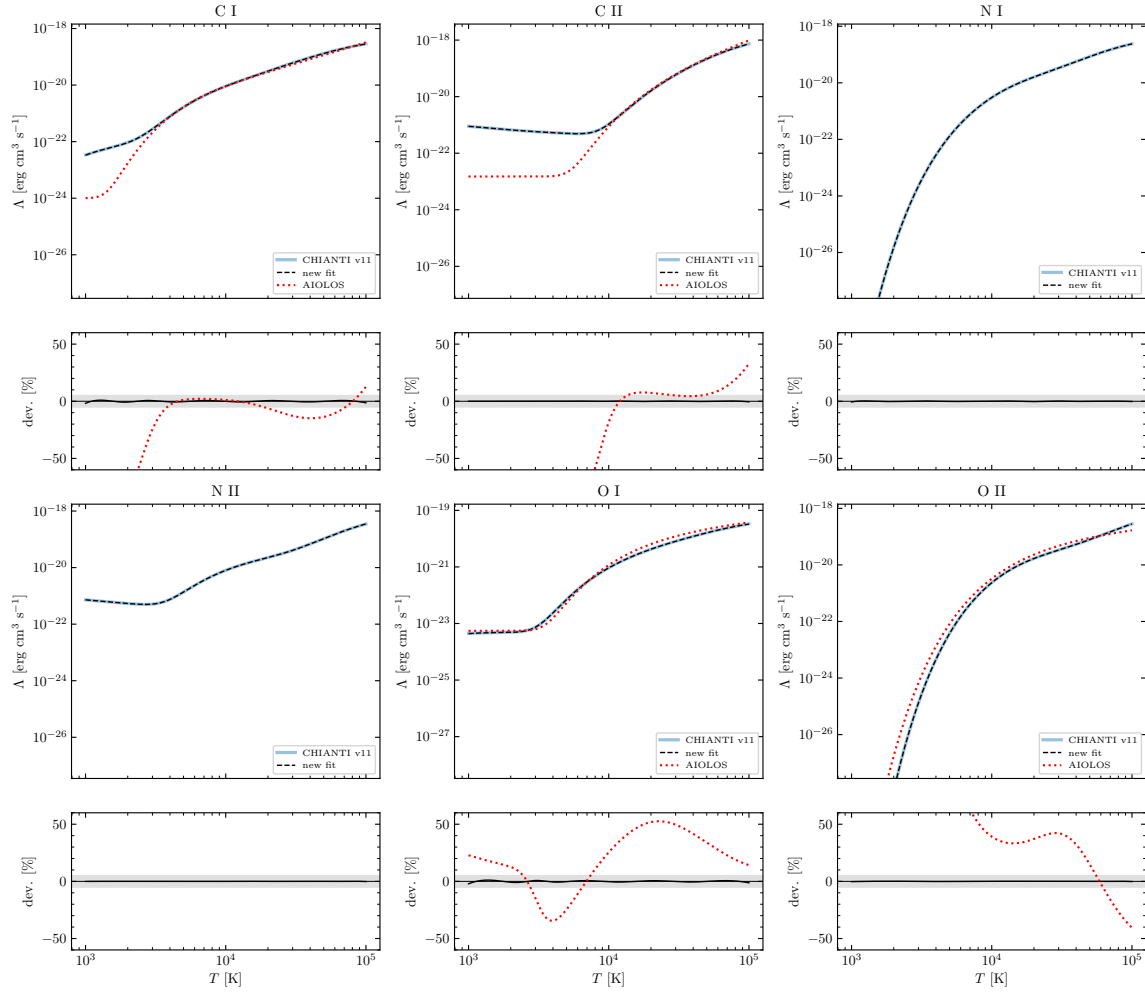


Figure 3: C/N/O line cooling: CHANTI v11 (solid blue), the new closed-form fits (black dashed), and the legacy AIOLOS formulas (red dotted; absent for nitrogen). Deviation strips show the percentage difference from CHANTI; the grey band is $\pm 5\%$.

base density. In the CHIANTI mode the floor is therefore replaced by the exact two-level solution

$$W_{\text{FS}}(T, n_e, n_{\text{HI}}) = f_l(T) \frac{h\nu A_{ul} x C_{\text{dex}}}{A_{ul} + C_{\text{dex}}(1+x)}, \quad x = \frac{g_u}{g_l} e^{-E/k_B T}, \quad C_{\text{dex}} = n_e k_e(T) + n_{\text{HI}} k_{\text{H}}(T), \quad (5)$$

in erg s^{-1} per ion, plus a multi-exponential refit of the CHIANTI curve with the $1 \rightarrow 2$ channel removed (the “remainder”). The ground-term Boltzmann weight $f_l = g_l/Z_{\text{term}}(T)$ makes the high-density limit of Eq. (5) equal to the exact multi-level LTE population result, while the $C_{\text{dex}} \rightarrow 0$ limit reproduces the CHIANTI coronal channel by construction (verified to $\leq 3.4\%$, set by the cubic log- Υ fits below). The assembly uses $\Lambda_{\text{eff}} = W_{\text{FS}}/n_e + \Lambda_{\text{rem}}(T)$, so the n_e prefactor cancels and the H-collision channel — which the electron-only coronal curves lack entirely — is recovered exactly.

Atomic data: [C II] $E/k_B = 91.21$ K, $A_{ul} = 2.290 \times 10^{-6} \text{ s}^{-1}$, $g_l=2$, $g_u=4$ (the two-level form is exact); [O I] $63 \mu\text{m}$ $E/k_B = 227.71$ K, $A_{ul} = 8.542 \times 10^{-5} \text{ s}^{-1}$, $g_l=5$, $g_u=3$, with $f_l = 5/(5+3e^{-227.7/T} + e^{-326.6/T})$. The electron de-excitation rate is $k_e = 8.629 \times 10^{-6} \Upsilon(T)/(g_u \sqrt{T})$ with $\log_{10} \Upsilon$ a cubic in $x = \log_{10}(T/10^4)$ fitted to the CHIANTI Υ (max err 3.4–3.6%); the H-atom rates follow the values used by the legacy `use_2lev_cool` branch (4.0×10^{-11} for C II, Goldsmith et al. 2012; $4.2 \times 10^{-11} (T/100)^{0.67}$ for O I, Draine 2011 / Lique et al. 2017; both flagged approximate). The weaker [O I] $146 \mu\text{m}$ and the C I / N II fine-structure floors remain coronal — each is a few-percent piece of its floor.

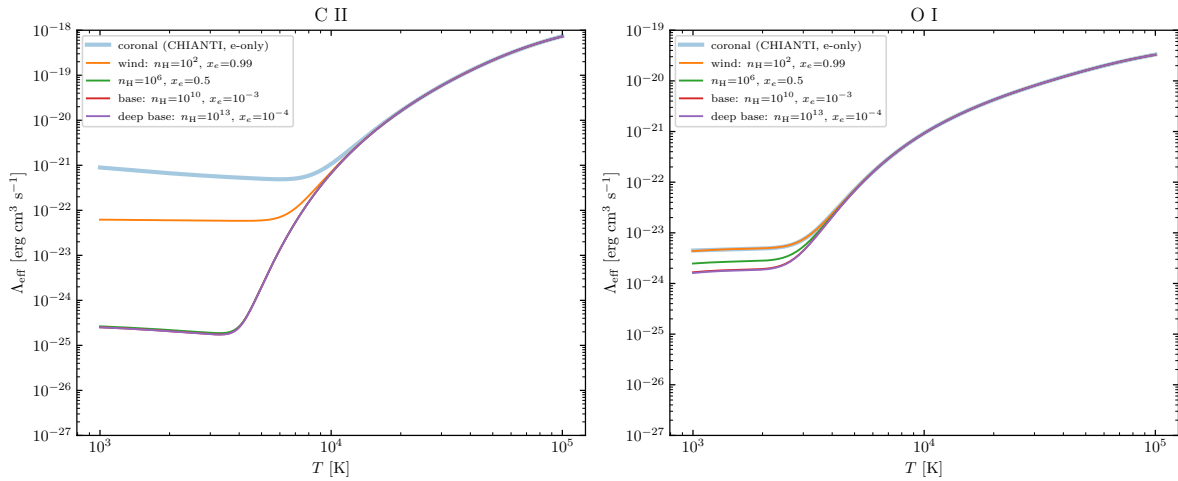


Figure 4: Effective C II (left) and O I (right) cooling coefficients with the fine-structure floor saturated, Eq. (5), for densities from the wind to the deep base, compared with the electron-only coronal curve. The [C II] floor is suppressed by up to $\sim 4 \times 10^3$ at base densities; for O I the H-collision channel partly offsets the saturation in mostly neutral gas.

The practical impact is large: on WASP-121b the *unsaturated* CHIANTI floors had lowered the base temperature by ~ 1000 K and the mass-loss rate by 15% ($\log \dot{M}$ 13.31 $\rightarrow$ 13.24); with Eq. (5) both effects disappear ($\log \dot{M} = 13.31$; saturated vs. AIOLOS profiles differ by at most 111 K, 1.7%). This appears to indicate that the small constant floors of the AIOLOS fits, Eq. (4), effectively encode the saturated regime, which is likely why they performed well in practice. The genuinely new physics of the CHIANTI mode is then the nitrogen cooling, the corrected O I/O II curves, and the H-collision fine-structure channel.

6. Free–free and the remaining tabulated pieces

Free–free cooling keeps the Huang (2023) Eq. (9) form, $\Lambda_{\text{ff}} = 1.9095 \times 10^{-25} Z^2 (T/10^4)^{0.55}$ per $(n_e n_{\text{ion}})$. Two coefficients remain tabulated because they are not reducible to closed form: Fe I (NIST f -values + Van Regemorter over the even-parity metastable manifold; 1-D in T) and the Fe II multilevel statistical-equilibrium coefficient $\Lambda_{\text{eff}}(T, n_e)$ (2-D). A possible future step, analogous to Sect. 5., would be a closed-form saturation treatment of the Fe II a^6D manifold, but the 2-D table is exact within its grid and carries no runtime cost concern.

Known limitations. (i) The legacy `use_2lev_cool` branch hard-codes the AIOLOS exponential parts for C II/O I and is not reached by the CHIANTI formulas; it is superseded by Sect. 5. and kept only for comparison (default off). (ii) The H-collision rates in Eq. (5) are order-of-magnitude literature values, not fits to a specific calculation (!To Be Checked in the source). (iii) All formulas are calibrated over 10^3 – 10^5 K; outside this range they extrapolate smoothly but are unvalidated.